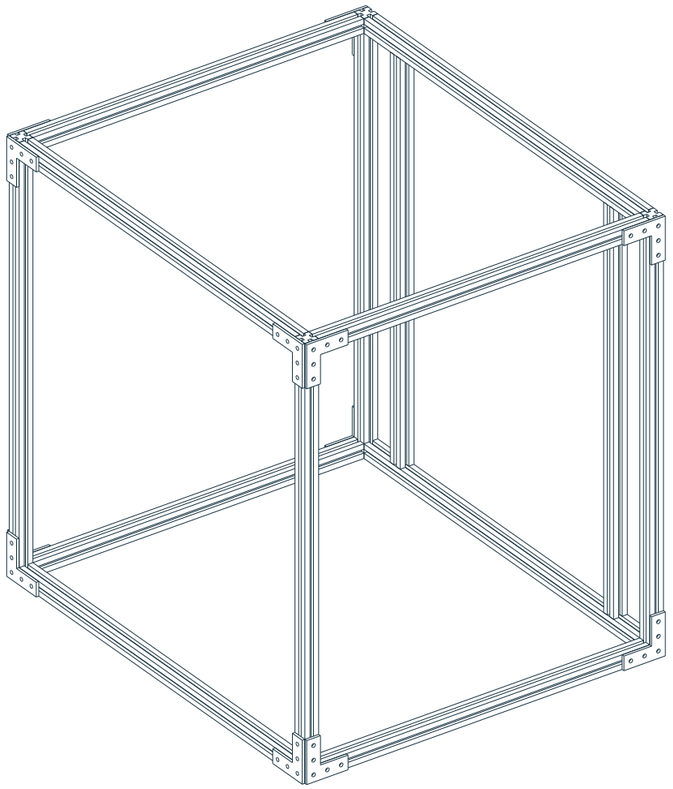
Rev C / Purchased-part architecture

A traceable development review, separate from the preserved Rev A and Rev B packages



The structure view omits door, panels, base brackets, fasteners, phone fixture and cabinet. No phone dimensions are assumed.

1010 aluminum structure
762 x 762 mm footprint. A stock metric breadboard carries a complete rotary stage with a vertical axis.

What this package establishes
Original CAD provenance, structure placements, measured mounting patterns and two rejected fastener lengths.

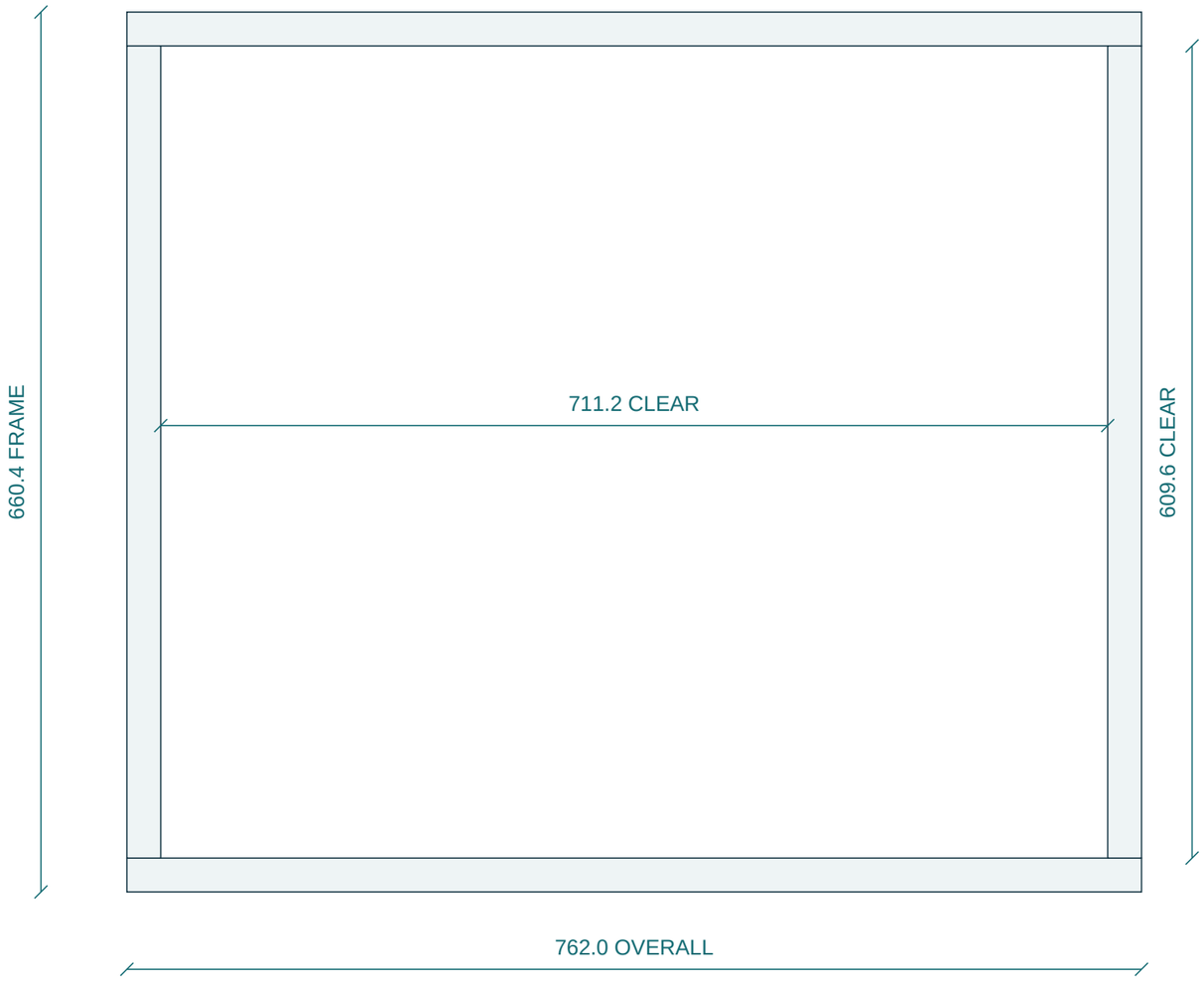
Full closure is required
Back-surface vacuum grip, passive compliance, force protection, guard details and complete electrical wiring.

No fabrication permission
Buy the catalog components. These reconstructed views are not drawings from which to manufacture substitutes.

Evidence inventory
46 pinned vendor files. The structure STEP contains 28 instances; omitted hardware is listed explicitly.

Structure envelope and cut schedule

Proposed assembly dimensions; supplier cuts only; all lengths in millimetres



ID	Qty	Length	Location
C101	4	762.0	Front/rear, base and roof
C102	4	711.2	Side rails, base and roof
C103	4	609.6	Four vertical posts
C104	2	711.2	Breadboard supports at Y +/-275

Order 80/20 1010, square cuts, no end taps or access holes. Door profiles are not included in this schedule.

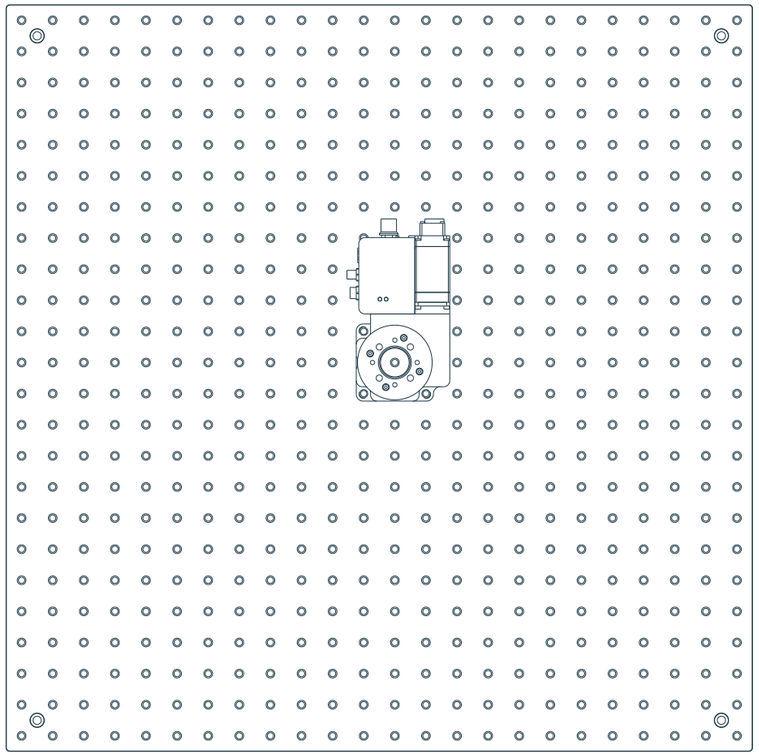
Coordinate origin: center of the footprint, bench surface Z=0. Base rail centers Z=12.7; roof rail centers Z=647.7; posts span Z=25.4 to 635.0.

C105 exterior plates use five C106 screw/T-nut assemblies each. Eight plates join posts in front/rear planes; four join roof corners. Base corner/support brackets remain to be modeled.

Slot interiors in the CAD are indicative. Panel capture, fastener bottoming, frame stiffness and door clearances require closure before release.

Purchased breadboard and stage placement

Original Thorlabs and Zaber STEP geometry; plan view; no fabricated base plate



Interface	Design value
Breadboard	MB6060/M; 600 x 600 x 12.7
Grid	M6 x 1; 25 mm spacing; 12.5 mm edge offset
Board bottom / top Z	25.4 / 38.1
Rotary axis X, Y	12.5, 12.5 relative to board center
Stage support Z	38.1
Board fasteners	Four corner counterbores; supports at Y +/-275

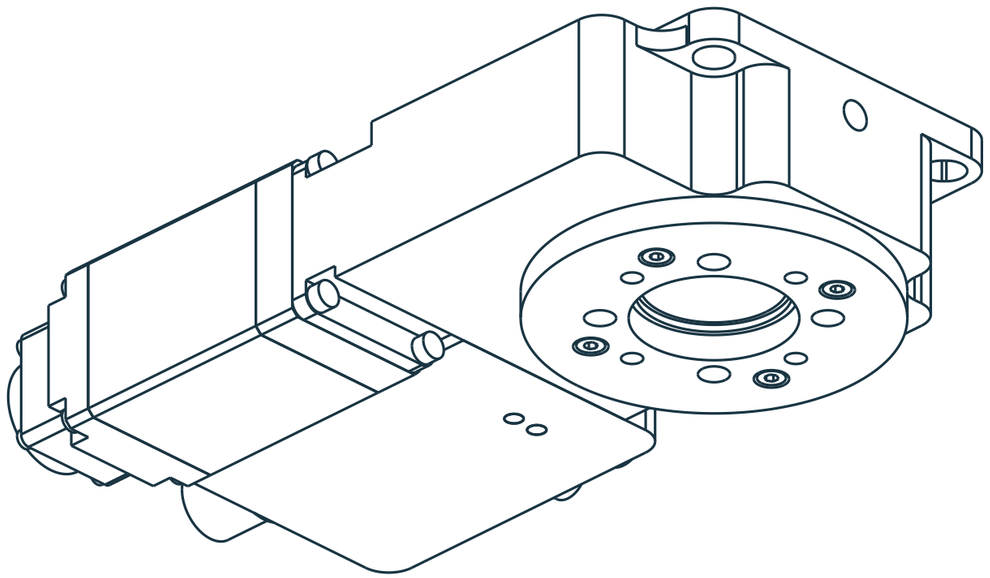
Three stage base screws

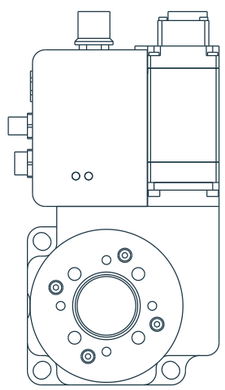
In the modeled orientation, breadboard coordinates are (-12.5, -12.5), (-12.5, 37.5), and (37.5, -12.5). The fourth quadrant is not a fourth mounting hole.

The base mounting CAD has 5.5 mm seating height. Screw lengths and washers for the breadboard and its 10-series T-nuts are still open; do not reuse Rev B hardware by assumption.

Rotary stage / catalog interface

Zaber X-RSW60A-E03 - original CAD; model placement is separate from functional qualification





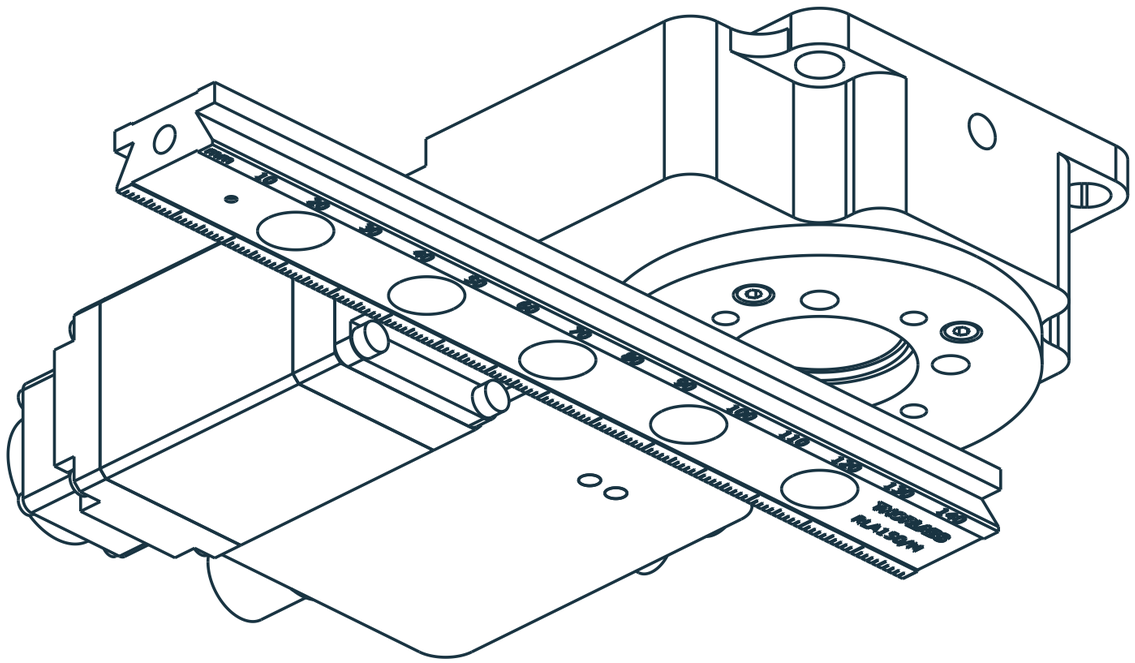
Feature	Manufacturer data / CAD observation
Rotating top	Diameter 60 mm
Metric pattern	4 x M6 x 1 on 25 mm square; drawing depth 6.5
Imperial pattern	Separate #8-32 pattern; do not substitute
Height	31.75 CAD; 31.7 rounded drawing
Supply / link	24-48 V; RS232 ASCII; M8 power path also present
Encoder	Motor-side measurement; not phone angle sensing

Design hold C3

The published 2.25 N m continuous-torque capability is not an allowable phone load. Current limiting does not establish hinge torque. Force protection and safe release of retained load remain unresolved.

Rail mounting and blind-thread engagement

Candidate interface only - original CAD stack; inspect received parts before selecting screws



Check	Result
Rail	RLA150/M; 5 mounting positions at 25 mm pitch
Stage row	Two adjacent metric holes; offset 12.5 from axis
Seat stack	2.54 nominal from STEP support plane to screw seat
M6 x 8 candidate	8 - 2.54 = 5.46 nominal projection
M6 x 10 rejected	10 - 2.54 = 7.46; exceeds drawing depth 6.5

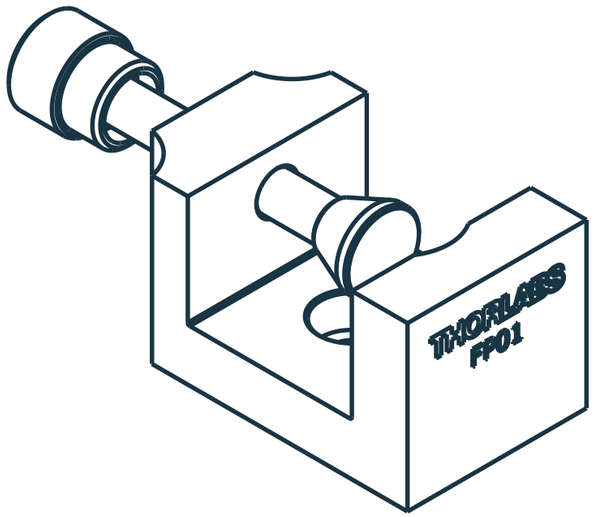
The modeled top holes are 5.75 mm deep; the vendor drawing calls out 6.5 mm. Treat this as a source discrepancy. Verify actual usable thread and screw clearance; do not lengthen screws based only on the drawing.

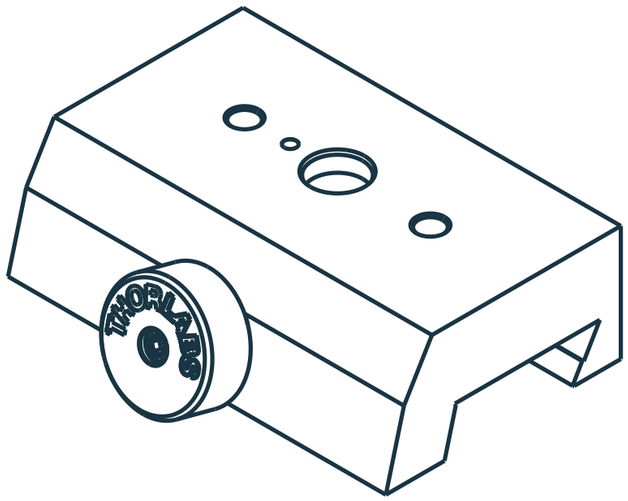
The offset rail does not place the phone hinge on the rotary axis. Transverse adjustment, carrier retention and acceptable drag are open interfaces.

This STEP study shows only stage + rail. It contains no fitted phone, passive slide qualification, end stops or released fastener assembly.

Rejected edge contacts / retained fit audit

Actual clamp and carrier geometry - no phone envelope or force limit is invented





Item / check	Finding
FP01	Single M4 counterbore; 22.9 wide; 14.7 capacity
RC2/M	2 M4 taps; 6 mm usable thread; measured pitch 24.9936
FP02 direct pair	25.4 mm clearance-hole pitch; not approved on RC2/M
FP01 + M4 x 8	Nominal 5.714 mm engagement; inspection pending
FP01 + M4 x 10	7.714 mm projection; rejected

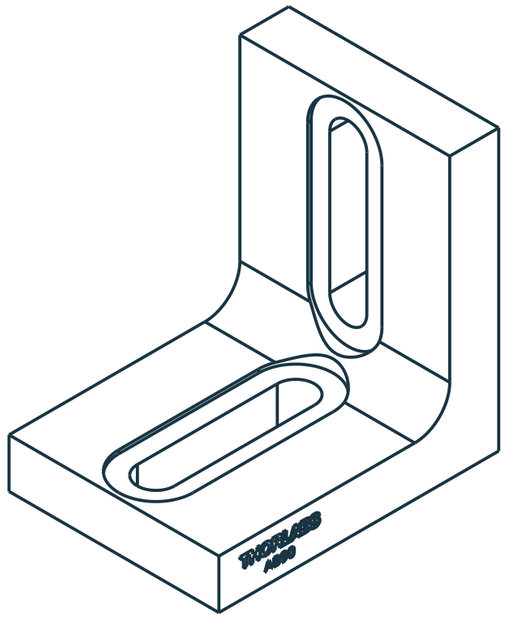
Owner decision: full closure required
Edge clamps are rejected for the working fixture. Their jaws occupy the closing gap. The replacement uses outside-surface vacuum contacts; qualification remains open.

Contact decision
Measure the actual phone. Identify rigid, permissible contact areas; exclude display, buttons, ports, cameras and hinge. No custom insert is permitted.

Compliance decision
The RC carrier is a spring-loaded dovetail. Measure breakaway and running drag under the intended load. A loose carrier knob is not a proven bearing or torque limiter.

Outside-surface vacuum contacts / catalog interfaces

Replacement direction for full closure; purchasing and phone-load qualification remain on hold



Candidate	Catalog interface
ESS-20-SN / 189294	20 mm cup; NBR; M6 x 1 plug
Cup drawing	L1 11.6; plug L2 2.4; suction stroke 1.8
ESH-HB-4-QS / 189209	M6 female mount; M6 cup seat
Holder drawing	H1 28; D7 12; side port QS-6
AB90	Stock M6 counterbored slots; adjustment by sliding
Quantity direction	2 cups per outside leaf; 4 total

Interface chain
Purchased support - AB90 - M6 screw through slot - ESH holder - matching ESS cup. Screw length, usable holder thread and support connection remain unverified; no drilling or tapped adapters are allowed.

Contact limitation
The 189294 datasheet lists 16.3 N holding and 11.5 N lateral force at -0.7 bar. These are catalog values, not allowable phone loads, peel capacity or a certified grip on coated phone surfaces.

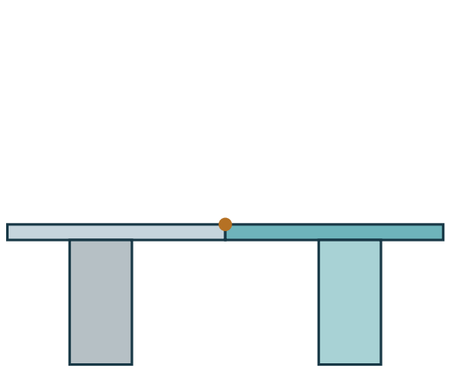
Vacuum system still required
Select a purchased generator, branch monitoring, hoses, filter and retained-grip stop/recovery arrangement. Do not release vacuum on a motor stop and let the phone fall. Cup slip is not a torque limiter.

AB90 - original Thorlabs CAD
50.8 mm legs; 38.1 mm width; nominal 8.9 mm wall. Both faces have a counterbored slot for M6 cap screws and washers. The drawing places slot arc centers 25.4 mm apart. This is a purchased bracket, not permission to machine a substitute.

Full closure / contact-envelope study

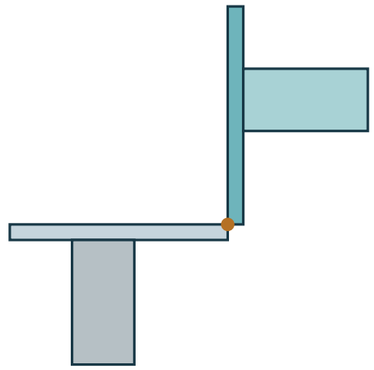
Owner requirement: full natural closure. Illustrative single-axis surrogate, not measured iPhone geometry

180 deg opening



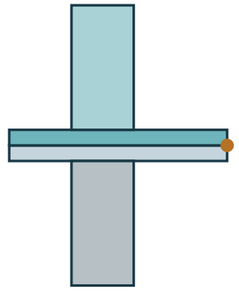
Outside grips stay out of gap

90 deg opening



Outside grips stay out of gap

0 deg opening



Outside grips stay out of gap

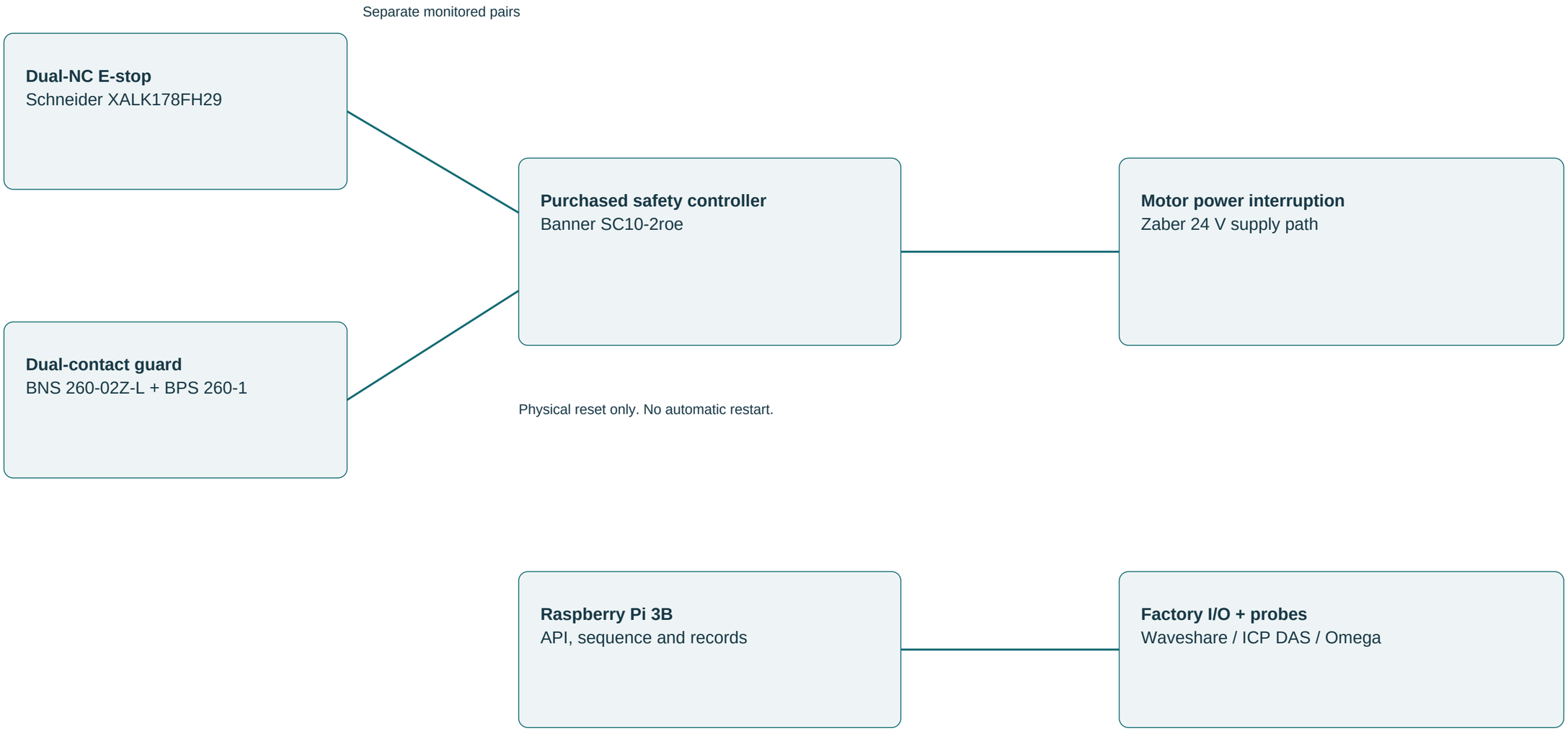
Study input	Value / boundary
Surrogate leaf	70 mm wide x 5 mm thick; not phone data
Grip envelope	20 mm wide x 40 mm deep; 40 mm radius
Sweep	361 positions at 0.5 degree increments
Result	No overlap of these four 2D envelopes
Not included	Supports, cables, hoses, real hinge path

Physical endpoint
Measure the natural closed state on the actual phone. An ideal diagram at zero opening angle does not authorize forcing a phone to zero degrees. Record hinge motion and clearance over the full path.

What this study establishes
Moving the contacts outside the leaves removes the edge-jaw obstruction in the simplified model. It does not establish that the complete stock mounting chain clears, retains the phone, or limits load. The separate phone measurement traveler defines the evidence needed before detailed fixture release.

Controls / independent stop architecture

Functional block diagram only - terminal wiring and safety-controller configuration are not released



Pi observes permit and commands motion; it does not replace the hardwired stop function.

Electrical interface checks

Purchasable modules replace Rev B's custom interface board; fit and wiring remain controlled holds

Component	Verified data	Remaining work
BNS 260-02Z-L	26 x 36 x 13; two 4.5 mounting holes / 22 pitch; no LED	Direct T-slot mounting, tamper resistance, gap and stop-time validation
SC10-2roe	45 x 115 x 100; 24 V; inputs 5 mA typical / 50 mA peak	Separate guard/E-stop inputs; physical reset; validated configuration
Current compatibility	Non-LED BNS 260: 400 mA maximum; SC10 peak 50 mA	Rating comparison only; circuit validation still required
Waveshare 26244	72 x 121 x 34; DIN; 7-36 V; Modbus RTU	DI polarity, address, stale-data behavior and wiring
M-7018R	Manufacturer 76 x 120 x 42; Type K; 10-30 V	Require firmware supporting open-wire detection; verify maps
Omega SA1-K	Self-adhesive 13 x 19 x 0.5 pad; factory 1 m leads	Phone surface suitability, lead routing and response check
Enclosure / harness	Factory entries, stock rails, sealed external supplies	Complete cable-entry plan, mounts, fuse schedule and BOM

Guard candidate differs from the early 80/20 22202 OSSD proposal. Do not mix their wiring. The original 22202 datasheet is retained solely as alternative-source evidence.

Procurement status / working register

Not a complete purchase BOM - candidate quantities and unresolved accessories are intentional holds

ID	Qty	Vendor / part	Description	Status
C101	4	80/20 / 1010	762 mm supplier cut	Structure study
C102	4	80/20 / 1010	711.2 mm supplier cut	Structure study
C103	4	80/20 / 1010	609.6 mm supplier cut	Structure study
C104	2	80/20 / 1010	711.2 mm breadboard supports	Structure study
C105	12	80/20 / 4151	76.2 x 76.2; 4.7752 thick	Modeled interface
C106	60	80/20 / 3321	5 screw/T-nut sets per 4151	Not modeled
C107	8	80/20 / 4119	Base corners/support connections	Placement review
C108	16	80/20 / 3061 + 3382	2 screw/T-nut pairs per 4119	Placement review
C109	4	80/20 / 3873	M6 nut for breadboard; screw TBD	Stack open
C110	1	Thorlabs / MB6060/M	600 x 600 x 12.7; M6/25 grid	Original STEP
C111	1	Zaber / X-RSW60A-E03	60 mm rotary top; 24-48 V	Original STEP
C112	candidate	Thorlabs / RLA150/M	150 mm dovetail rail	Top interface only
C113	candidate	Thorlabs / RC2/M	M4 taps; 24.9936 mm CAD pitch	Drag/retention open
C114	candidate	Thorlabs / RC1	25.4 mm carrier as stop	Retention open
C115	0	Thorlabs / FP01	22.9 mm wide; 14.7 mm capacity	REJECTED: closure gap
C116	candidate	Thorlabs / DT12/M	12.7 mm manual adjustment	Mounting open

Procurement status / working register

Not a complete purchase BOM - candidate quantities and unresolved accessories are intentional holds

ID	Qty	Vendor / part	Description	Status
C120	candidate	80/20 / 2086	10-series metal hinge	Door not released
C121	candidate	80/20 / 2643	3 mm clear polycarbonate	Panel sizes open
C122	candidate	80/20 / 2116	10-series panel gasket	Capture open
C201	1	Banner / SC10-2roe / 806222	45 W x 115 H x 100 D	Configuration open
C202	1	Schmersal / BNS 260-02Z-L	101184385; 26 x 36 x 13	Mounting open
C203	1	Schmersal / BPS 260-1	101184395; matching actuator	Mounting open
C204	4	80/20 / 3666	10-series M4 roll-in T-nut	Mounting open
C205	1	Schneider / XALK178FH29	Dual-NC complete E-stop station	Wiring/mount open
C206	1	Waveshare / 26244	72 x 121 x 34; Modbus RTU I/O	I/O map open
C207	1	ICP DAS / M-7018R	76 x 120 x 42; thermocouple input	Firmware/map open
C208	1 pack	Omega / SA1-K	5-pack; use 2; 13 x 19 x 0.5 pad	Attachment open
C209	1	ICP DAS / tM-7561	Isolated USB to RS485	Bus design open
C210	1	Raspberry Pi / 3 Model B	Headless host; separate supply	Mount/case open
C150	4 proposed	Festo / ESS-20-SN / 189294	20 mm flat NBR cup; M6 vacuum thread	Contact qualification open
C151	4 proposed	Festo / ESH-HB-4-QS / 189209	28 mm holder; M6 mount; 6 mm tube port	Mount screw stack open
C152	2 proposed	Thorlabs / AB90	50.8 x 50.8 x 38.1 angle; counterbored M6 slots	Stock mounting candidate

Assembly release gates

The next assembly traveler must close these items; none are delegated to undocumented shop fabrication

Gate	Required evidence before release
C1 / Phone contact	Full natural closure; actual phone measurements; purchased grip; stock mounting chain; no display loading.
C2 / Compliance	Measured hinge path; retained slide travel; breakaway/running drag; cyclic wear and contamination behavior.
C3 / Force protection	Permitted phone loads; qualified independent limiting/protection; retained-load recovery procedure.
C4 / Structure and guard	Fasteners and brackets; door/panel geometry; sensor mounting; door gaps, stopping time and reach assessment.
C5 / Electrical	Factory enclosure entries; detailed wire/terminal schedule; fuses; configuration record; supply load budget.
C6 / Software	Rev C drivers and OpenAPI; communication/watchdog faults; temperature open-wire and staleness; explicit local commissioning.
C7 / Handoff	Complete BOM; separate assembly traveler; full CAD interference review; documented physical acceptance.

The Rev B application uses a different actuator and I/O architecture. It must not be installed as a working Rev C hardware controller. Simulation and preserved historic viewers remain available independently.

Source provenance and rebuild record

Every original download is pinned by SHA-256; original vendor ownership is retained

46 original reference files are indexed in hardware/sources.json. The JSON records exact URLs, retrieval date, byte count and checksum. Original files are retrieved with scripts/fetch_sources.py; --verify-only checks existing copies. The public repository contains the source register and our derived work, not a blanket redistribution of vendor manuals.

Primary evidence	Original source
Zaber stage	zaber.com - X-RSW60A-E03 STEP and X-RSW-E drawing
Thorlabs mechanism	media.thorlabs.com - original part STEP and PDF assets
Banner controls	info.bannerengineering.com - 174868 manual; 209332 drawings
Schmersal guard	products.schmersal.com - BNS 260 operating instructions
I/O and probes	icpdas.com - M-7018R data; br.omega.com - SA1 data
1010 structure	8020.net - 1010, 4151, 3321, 4119 and mounting hardware
Vacuum contacts	ftp.festo.com - ESS and ESH drawings; 189294/189209 data

Rebuild sequence
Use the repository Pipenv environment: fetch_sources.py, check_interfaces.py, build_structure.py, build_closure_study.py and build_review.py, all under revisions/rev-c/scripts. No new production dependency is required.

Measured audit
CAD interface checks passed; NOT build release. Five hole-pattern checks pass. This is evidence about CAD interfaces, not proof of mechanical function, fatigue life, phone compatibility or machine safety.